EQUIVALENCES OF SOBRIETY, UNIQUENESS OF CLOSED POINTS, AND THE IRREDUCIBLE-OPEN CRITERION FOR REASONABLE SPACES

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Sobriety transfer. Properties of Spaces, lemma points-cartesian and section-references on sober schemes (cross-chapter, named). Schemes are sober; sobriety pulls back through etale covers.
- No specializations within fibres. lemma-no-specializations-map-to-same-point (carded L26707). The collapse used in the irreducible direction.

Source reference. Original lemma 03K3: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2893–2893. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03KF: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1126–1126. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01IS: focused-seeds/geometry/S9-decent-reasonable-sober/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IQ: focused-seeds/geometry/S9-decent-reasonable-sober/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 086U: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2323–2323. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 004X: focused-seeds/geometry/S9-decent-reasonable-sober/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof

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cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Then the topological space $|X|$ is sober (see Topology, Definition 004X).*

Proof. We have seen in Lemma 03K3 that $|X|$ is Kolmogorov. Hence it remains to show that every irreducible closed subset $T \subset |X|$ has a generic point. By Properties of Spaces, Lemma 03IQ there exists a closed subspace $Z \subset X$ with $|Z| = |T|$. By definition this means that $Z \to X$ is a representable morphism of algebraic spaces. Hence Z is a decent algebraic space by Lemma 03KF. By Theorem 086U we see that there exists an open dense subspace $Z' \subset Z$ which is a scheme. This means that $|Z'| \subset T$ is open dense. Hence the topological space $|Z'|$ is irreducible, which means that Z' is an irreducible scheme. By Schemes, Lemma 01IS we conclude that $|Z'|$ is the closure of a single point $\eta \in T$ and hence also $T = \overline{\{\eta\}}$, and we win. $\square$